3

PROCESS CONTROL VS. SAFETY CONTROL

Chapter Highlights

3.1 Control and Safety Defined
3.2 Process Control - Active/Dynamic
  3.2.1 The Need For (and Ease of) Making Frequent Changes
3.3 Safety Control - Passive/Dormant
  3.3.1 The Need for Restricting Changes
  3.3.2 Demand Mode vs. Continuous Mode
3.4 Separation of Control and Safety Systems
  3.4.1 HSE
  3.4.2 AIChE CCPS
  3.4.3 IEC 61508
  3.4.4 ISA SP84
  3.4.5 API RP 14C
  3.4.6 API RP 554
  3.4.7 NFPA 85
  3.4.8 IEEE 803
3.5 Common Cause and Systematic/Functional Failures
  3.5.1 Human Issues
Summary
References

"Nothing can go wrong, click… go wrong, click… go wrong, click…"

— Anonymous

Process control used to be performed in pneumatic, analog, single loop controllers. Safety functions were performed in different hardware, typically hardwired relay systems. Electronic distributed control systems (DCSs) started to replace single loop controllers in the 1970s. Programmable logic controllers (PLCs) were developed to replace relays in the late 1960s. Since both systems are software programmable, some people naturally concluded that there would be benefits in performing both control and safety functions within the same system, usually the DCS. The typical benefits touted included single source of supply, integrated communications, reduced training and spares, simpler maintenance, and potentially



lower overall costs. Some believe that the reliability, as well as the redundancy, of modern DCSs are "good enough" to allow such combined operation. However, all domestic and international standards, guidelines, and recommended practices clearly recommend *separation* of the two systems. The authors agree with this recommendation and wish to stress that *the reliability of the DCS is not the issue*.

3.1 Control and Safety Defined

Critical systems require testing and thorough documentation. It's debatable whether normal process control systems require the same rigor of testing and documentation. When the US government came out with their process safety management (PSM) regulation (29 CFR 1910.119) in 1992, many questioned whether the mandated requirements for documentation and testing applied to both the control systems as well as the safety systems. For example, most organizations have documented testing procedures for their safety instrumented systems, but the same may not be said for all of their control system loops. Users in the process industry questioned OSHA representatives as to whether the requirements outlined in the PSM regulation applied to all 6,000 loops in their DCS, or just the 300 in their safety instrumented systems. OSHA's response was that it

included everything. Users felt this was another nail in the proverbial coffin trying to put them out of business.

This helped fuel the development of ANSI/ISA-91.01-1995 "Identification of Emergency Shutdown Systems and Controls That Are Critical to Maintaining Safety in Process Industries [Ref. 10]". The ANSI/ISA-91.00.01-2001 standard was reaffirmed in 2001. This brief standard (only two pages long) includes definitions of process control, safety control, and safety critical control.

The ANSI/ISA-91.00.01-2001 standard defines a "Basic Process Control System" (BPCS), as opposed to an 'advanced' process control system, as "the control equipment installed to perform the normal regulatory functions for the process (e.g., PID control and sequential control)." Some have stated that this accounts for up to 95% of instrumentation for most land-based facilities. Most people accomplish this with a DCS, PLC, or hybrid system.

The ANSI/ISA-91.00.01-2001 standard defines "Emergency Shutdown System" as "instrumentation and controls installed for the purpose of taking the process, or specific equipment in the process, to a safe state. This does not include instrumentation and controls installed for non-emergency shutdowns or routine operations." Some have stated that this accounts for less than 10% of instrumentation for most land-based facilities.

The ANSI/ISA-91.00.01-2001 standard also defines a third category, "Safety Critical Control," as "a control whose failure to operate properly will directly result in a catastrophic release of toxic, reactive, flammable or explosive chemical." This essentially means any control or safety layer, without additional safety layers, where a failure will immediately result in a hazardous event. Hopefully, the percentage of such loops is very small.

Based on these definitions, users stated to OSHA that their 6,000 DCS loops were not safety-related and therefore did not require the same degree of rigor for documentation and testing as their 300 safety instrumented loops.

This is not meant to imply that the design of distributed control systems does not require thorough analysis, documentation, and management controls. They obviously do, just not to the same extent as safety systems.

3.2 Process Control - Active/Dynamic

It's important to realize and understand the fundamental differences between process control and safety control. Process control systems are active, or dynamic. They have analog inputs, analog outputs, perform math and number crunching, and have feedback loops. Therefore, most failures in control systems are inherently self-revealing. For example, consider the case of a robot on an automated production line. Normally the robot picks up part A and places it in area B. If the system fails, it's obvious to everyone; it no longer places part A in area B. There's no such thing as a "hidden" failure. The system either works or it doesn't. There's only one failure mode with such systems—revealed—and you don't need extensive diagnostics to annunciate such failures. If a modulating process control valve were to fail full open or full closed, it will most likely impact production and the problem would become evident to everyone very quickly. If the valve were to freeze in its last state the resulting problems would also make themselves known relatively quickly. Again, extensive testing and diagnostics are usually not required to reveal such failures or problems.

3.2.1 The Need for (and Ease of) Making Frequent Changes

Process control systems must be flexible enough to allow frequent changes. Process parameters (e.g., set points, PID settings, manual/automatic, etc.) require changing during normal operation. Portions of the system may also be placed in bypass and the process may be controlled manually. This is all normal for control systems. Simple, easy, and quick access is a requirement in order to make such changes.

3.3 Safety Control - Passive/Dormant

Safety systems, however, are just the opposite. They're dormant, or passive. They operate for extended periods of time doing virtually nothing and hopefully will never be called into action. An example would be a pressure relief valve. Normally, the valve is closed. It only opens when the pressure reaches a certain limit. If the pressure never exceeds that value, the valve never operates. Many failures in these systems may *not* be self-revealing. If the relief valve is plugged there is no immediate indication. A PLC could be hung up in an endless loop. Without a watchdog timer, the system would not be able to recognize the problem, and if the system was dormant, neither would you. An output module might use triacs which could fail energized. Many systems are unable to recognize such a prob-

lem. If the output was energized during normal operation, how would you recognize such a condition? How confident are you that a valve that hasn't been stroked in seven years will actually close? How confident are you that a backup generator that hasn't even been turned on in two years will actually run when you need it? For that matter, how confident are you that your lawn mower will start if it's been left sitting in the garage all winter long with gasoline in the tank and carburetor? How much confidence can you place in anything that under normal circumstances is not functioning (or changing state) to begin with? The only way to know if dormant systems will work when you need them is to test them. Either you must test the system or it must be able to test itself. Hence, there is a need for extensive diagnostics in dormant, passive safety-related systems. The alternative is to use inherently fail-safe systems, where such dangerous failures are highly unlikely by design, but this is generally not possible for all components.

What this means is that systems designed for *control* may be totally unsuitable for *safety*. The two primary issues have to do with security (i.e., access control) and diagnostics.

Safety systems require a high level of access control in order to implement security features and enforce management of change procedures. Safety functions should not be tampered with. How do you provide operators with easy access to some functions, but lock them out of others? How do you know that making a change in one area of the system won't impact a critical function elsewhere? Many control systems do not implement very effective access control measures.

Extensive diagnostics are required in safety systems simply because not all failures are self-revealing. Many people *assume* that all modern electronic systems include extensive diagnostics. Unfortunately, this is *not* the case. There are very simple economic reasons why this is so. This will be covered in more detail in Chapter 7.

3.3.1 The Need for Restricting Changes

Safety instrumented systems should be designed to allow very little human interaction. Operators interact with and control the process using the control system. If control systems can no longer maintain control, independent alarms generally indicate a problem. Operators can then supervise manual operations. If automatic process control systems and human intervention cannot make the required corrections, the last line of defense should function automatically and independently. About the only interaction that should be allowed between operators and safety systems

are overrides for allowing startup and maintenance work on portions of the system. Note that strict procedures must be adhered to at such times. People must know that there is a last line of defense—something that can be relied upon. These systems should have tightly controlled access. The last thing anyone wants is a system that doesn't function because some lone operator disabled the system without anyone else's knowledge. It has happened.

3.3.2 Demand Mode vs. Continuous Mode

The 1993 AIChE CCPS guideline text [Ref. 3] and the 1996 version of ANSI/ISA-84.01 [Ref. 5] considered safety systems to be low demand mode systems, although this term wasn't used at that time. The thought was that properly designed safety systems would have demands placed on them infrequently (e.g., once per year). The more recent IEC 61508 standard [Ref. 4], which was written to address *all* industries, introduced the concept of high demand—or continuous mode—systems. These are simply systems that have frequent demands placed on them. A classic example is the brakes in a car. The concept of continuous mode systems filtered down into the IEC 61511 standard for the process industry and the ISA SP84 committee.

As one might expect from any change, problems ensued. The safety system performance requirements originally documented for the different safety integrity levels (SILs), only apply for low demand mode systems. The concept of a "probability of failure on demand" does not make sense for systems that might have 10 or 10,000 demands placed on them in a year. While there are naturally some that disagree with the solution chosen, the committee adopted the concept of assigning dangerous failure *rates,* rather than probabilities, to the different SILs. The difference between the rates and probabilities is a factor of 10,000, based on the simplification that there are approximately 10,000 hours in a year.

Continuous mode systems are essentially similar in concept to what the ANSI/ISA-91.01 standard [Ref. 12] refers to as "safety critical control." Even members of the ISA SP84 committee have difficulty citing examples of continuous mode systems in the process industry. Some examples, such as a high level shutoff in a batch reactor that is loaded once per day, are considered by some people to be merely examples of poor process control, not safety.

3.4 Separation of Control and Safety Systems

An ongoing topic of controversy in the process industry is whether control and safety systems should be combined within one system. In other words, should all safety functions be combined in the process control system? Proponents would argue that both systems nowadays are programmable and that process control systems are reliable and can be redundant, so why not? The rebuttals are rather simple and do not hinge on reliability. As we shall see, all of the standards, recommended practices, and guidelines in industry recommend that *separate* systems be provided for process and safety control.

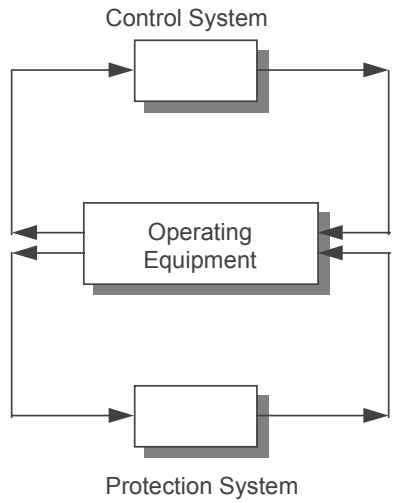
Proponents of combining control and safety also cite that since control systems are active and dynamic, and faults are inherently self-revealing (as stated earlier), that they're therefore ideally suited for use in safety since there would be no hidden faults. For example, an analog control valve is continually being stroked, which would identify problems. True, but not *all* problems. Having a separate solenoid on a control valve (powered by a separate safety system) may still not meet the overall safety requirements. Stroking a valve does not ensure that it will close or seal completely, and stroking the valve does not test that the solenoid will work when required. If the valve does fail stuck (everything fails, it's just a matter of when), then no safety action will be possible. Careful analysis is required to see if the safety requirements will be fulfilled when combining control and safety, as there are many other issues to consider.

Trevor Kletz, one of the industry leaders in terms of process safety has stated it simply and very well. "Safety systems such as emergency trips should be completely independent of the control system and, when practicable, hard-wired. If they are based on a computer, it should be independent of the control computer." [Ref. 1]

The following subsections describe what is stated in other documents.

3.4.1 HSE - PES

The English couldn't make the message of separation any more clear. Their document [Ref. 2] comes as a two part volume. Part I is for the managers and is only 17 pages long. Figure 3-1 takes an *entire page*. (The technical document—the one for the engineers—Part II, is about 170 pages.) Separate sensors, separate logic box, separate valves.



"It is strongly recommended that separate control and protection systems are provided."

Figure 3-1: U.K. Health & Safety Executive Recommendations

3.4.2 AIChE - CCPS

The American Institute of Chemical Engineers (AIChE) text [Ref. 3] is a set of "guidelines." About 50 pages of the text are devoted to what are termed interlock systems with additional good background material. Being a guideline, the text does not use words that imply mandatory requirements such as "shall." Instead, there are statements like 'normally...' that are always open to interpretation. However, some statements, like the ones below, are very straightforward.

"Normally, the logic solver(s) are separated from similar components in the Basic Process Control System (BPCS). Furthermore, SIS input sensors and final control elements are generally separate from similar components in the BPCS." What basis could you put forth for not following what a textbook written by industry experts—all end users at that—considered 'normal' and 'generally' appropriate?

"Provide physical and functional separation and identification among the BPCS and SIS sensors, actuators, logic solvers, I/O modules, and chassis..." In other words, physically separate the components so a failure of one will not impact the other. Implement different logic, because if the same logic is performed in both systems and there is an error in the functional specification, both systems will have the same error. Identify the

systems differently on drawings and out in the field so people realize that, "Oh, this is a safety device. I need to be careful and follow additional procedures."

3.4.3 IEC 61508

This international standard [Ref. 4] is for *all* industries and covers the use of relay, solid state, and programmable safety systems (or E/E/PES; electric, electronic, and programmable electronic systems).

"The EUC (equipment under control) control system shall be separate and independent from the E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities."

If this requirement cannot be met then the entire control system needs to be designed as a safety-related system and follow the requirements laid out in the standard. One such example would be aircraft controls. It's difficult to imagine a separate safety system designed to make an airplane at 35,000 feet fail to its safe state. In some cases control and safety are so intermingled that the entire system must be designed as safety-related.

3.4.4 ANSI/ISA-84.00.01-2004

The original 1996 version of this standard [Ref. 5] had the two following statements regarding separation. These statements were very black and white and easy to interpret.

"Sensors for SIS shall be separated from the sensors for the BPCS." Two exceptions were allowed: if there was sufficient redundancy, or if a hazard analysis determined there were enough other safety layers to provide adequate protection.

"The logic solver shall be separated from the BPCS…" There were exceptions for cases where it was not possible to separate control from safety functions, such as high speed rotating equipment like gas turbines.

ISA standards are generally revised every five years. Rather than completely rewrite the ISA SP84's standard, the committee agreed to adopt the IEC 61511 standard. The wording in the IEC standard regarding separation is not as clear-cut as in the 1996 version of the ISA SP84's standard. In fact, an appeal on this very issue delayed the release of the new standard an entire year. Users simply do not want separation *mandated* on them. Many believe that for very small systems, or systems with low performance requirements, combining control and safety may be acceptable,

assuming things are still done properly. The new standard states the following:

Clause 9.5.1 states, "The design…shall be assessed to ensure that the likelihood of common cause…and dependent failures between… protection layers and the BPCS are sufficiently low in comparison to the overall safety integrity requirements of the protection layers." The definition of 'sufficiently low' is naturally open to interpretation, as well as what constitutes a proper assessment.

Clause 9.4.3 states, "If a risk reduction greater than 10 is claimed for the BPCS, it shall be designed to the standard." Performance terms such as "risk reduction" will be described later in this text. What this clause means is that if you are going to combine control and safety in one box and claim a level of performance even in the lowest SIL 1 range, the system will need to be designed according to the standard. That might be feasible for a small system, but would be totally impractical for a large system.

Clause 11.2.4 states, "If it is intended not to qualify the basic process control system to this standard, then the basic process control system shall be designed to be separate and independent to the extent that the functional integrity of the safety instrumented system is not compromised." Again, statements such as "to the extent that…" are always open to interpretation.

Clause 11.2.10 states, "A device used to perform part of a safety instrumented function shall not be used for basic process control purposes, where a failure of that device results in a failure of the basic process control function which causes a demand on the safety instrumented function, unless an analysis has been carried out to confirm that the overall risk is acceptable." In other words, even field devices should be separate. However, there is still an "out" if "an analysis" can justify combining systems. What sort of justification? Simply stated, one that you would be willing to use in court.

3.4.5 API RP 14C

The American Petroleum Institute, Recommended Practice 14C "Analysis, Design, Installation & Testing of Basic Surface Safety Systems for Offshore Production Platforms [Ref. 6]" applies to the design of offshore platform shutdown systems. It prescribes how to determine the inputs and outputs for the shutdown system for different process units. It says nothing, however, on the design of the logic box. (It assumes most logic systems are pneumatic.)

"The safety system should provide two levels of protection... The two levels of protection should be independent of and in addition to the control devices used in normal operation."

It's surprising how open to interpretation some of these documents are. For example, would a relief valve and a gas detection system be considered "two levels of protection independent of and in addition to the control devices"? 14C tells you how to determine I/O and functional logic for the system. It doesn't tell you what sort of logic box to use. People have installed pneumatic, relay, PLC, and Triple Modular Redundant (TMR) systems on offshore platforms and *all* claim to have met RP14C, and indeed they may all have.

3.4.6 API RP 554

The American Petroleum Institute, Recommended Practice 554 "Process Instrumentation and Control [Ref. 7]" applies to refineries. Its statements regarding separation are rather straightforward. "The shutdown and control functions should be in separate and independent hardware… ," and "Shutdown devices should be dedicated and separate from pre-shutdown alarm and the process control system." In other words, safety-related logic systems and field devices should be dedicated and separate from control and alarm systems. Note that this document, as well as 14C, is a recommended practice, not a mandatory standard. It uses the word "should" rather than "shall".

3.4.7 NFPA 85

The National Fire Protection Association has written standards that apply to boiler and burner management systems, such as "Boiler & Combustion Systems Hazards Code [Ref. 8]." These might be viewed as examples of 'prescriptive' standards. Prescriptive standards tend to have black and white statements that are easy to interpret and, therefore, comforting to some. This standard makes a very clear call for separation. "Requirement for Independence: The logic system performing the safety functions for burner management shall not be combined with any other logic system." A DCS might be utilized for the 6,000 I/O associated with a huge industrial boiler, but the burner management system shall be a separate system. There is another statement that says, "The logic system shall be limited to one boiler only." In other words, using a redundant logic system capable of multiple programs to handle multiple burner management systems is not acceptable. However, facilities that are self-insured may not follow the NFPA standard.

3.4.8 IEEE 603

The Institute of Electric and Electronic Engineers has written standards that apply to the design of nuclear generating stations, such as "Standard Criteria for Safety Systems for Nuclear Power Generating Stations." [Ref. 9] The process industry is obviously not as restrictive as the nuclear industry, but the concept of separate safety layers is rather clear cut here.

"The safety system design shall be such that credible failures in and consequential actions by other systems shall not prevent the safety system from meeting the requirements."

Some nuclear facilities use a quad redundant computer arrangement, dual in series with dual in parallel. And as if that weren't enough, some used *two* of these quad redundant systems together! The U.S. typically uses quad systems backed up with 'conventional analog' systems.

An article in an English journal a few years back described the software testing for the programmable safety system of an English nuclear power station. It took 50 engineers 6 months to proof test the system! It's doubtful that many companies in the petrochemical industry could afford 25 man/years to proof test each of their safety systems! Someone with the Nuclear Regulatory Commission (NRC) told one of the authors that it wasn't 25 man-years, but more like several *hundred* man-years!

3.5 Common Cause and Systematic/Functional Failures

There are a variety of names and definitions for common cause. In essence, a common cause failure can be defined as a single stressor or failure that impacts multiple items or portions of a system.

One way of referring to such problems is the "Beta factor." This is the percentage of all identified failures in one "leg" or "slice" of a *redundant* system that might impact identical components and make the entire system fail. For example, suppose a central processor unit (CPU) has a failure rate of 10E-6 failures/hour. If the CPU were triplicated and all three were placed side by side in the same chassis, certain failures might impact two or three CPUs at once (e.g., a design, manufacturing, or software error). A Beta factor of 10% would mean that the system would act in a non-redundant manner with a failure rate of 1E-6 failures/hour. Note that the Beta factor model is purely empirical. It's not reality, just a model or estimation of reality. There are a number of documented techniques for estimating Beta value percentages. [Ref. 4 and 10]

Systematic failures, also called functional failures, impact an entire system and are therefore sometimes categorized as common cause failures. Examples of systematic failures usually include human errors in design, operation, and maintenance. However, heat, vibration, and other external factors can still be accounted for in the definition. Such failures may impact a redundant or non-redundant system. A systematic error in specifying a solenoid valve could result in the valve not operating properly at low temperatures. If redundant valves were used, the systematic failure would also be a common cause failure. While some systematic failures may be relatively easy to identify (e.g., a design error), the exact number or percentage (if accounting for them using the Beta factor) of such failures is usually more difficult to quantify. How accurately can one predict, for example, how often an engineering design error might occur, or that a maintenance technician might calibrate all three transmitters incorrectly?

One study in the nuclear industry found that 25% of all failures in nuclear power stations were related to common cause. [Ref. 10] Investigations in redundant control systems have found common cause percentages between 10% and 30%. It's surprising how some very redundant systems still have single points of failure. One networked computer systems had 7-way redundant communications. Unfortunately, they all ran through the same fiber optic cable which, when cut, disabled the entire system. [Ref. 11] Obviously no one would *intentionally* design a system this way. The point is, things eventually grow to the point where one individual can't "see" everything and certain items inevitably fall through cracks.

If control and safety functions are performed in the same system, there will always be potential for common cause faults. The more the systems are physically separated, the more unlikely it will be that single failures can affect them both. A simple phrase sums this thought up fairly well, "Don't put all your eggs in one basket." No matter how sturdy or reliable the basket may be, there will always be some unforeseen circumstances where it will drop. Everything fails, it's just a matter of when.

3.5.1 Human Issues

The various standards state that control and safety systems should be separate, but they don't really state *why*. The issues are actually rather straightforward. Much of it boils down to human issues.

Safety systems require strict management of change. If one were to combine control and safety in the same system, how effectively can you grant operators access to certain functions they need to change frequently, but lock them out of other functions that they should not? How assured can

you be that changes to one area of the system (e.g., software logic) do not impact a safety function elsewhere in the system? How assured can you be that placing a control loop in bypass, which is common practice, will not impact the operation of a safety alarm or action? The only way to have any real assurance is to have separate systems.

An actual example might help put this in perspective. This story was relayed to one of the authors during a lunch at one of the ISA SP84 committee meetings. A user company acquired an existing facility from another organization. An engineer from the parent company was sent to audit the control and safety systems at the acquired plant. The engineer found that all control and safety functions were performed in the DCS. Further checks were performed. It was found that one third of the safety functions had simply vanished (i.e., been deleted). Another third were found in bypass. The remaining third were tested and found not to be functional. One might make the argument that establishing and following proper procedures would have prevented such a situation. After all, procedures must be established for bypassing, maintaining, testing and management of change of safety systems. Unfortunately, procedures will often be violated, whether on purpose or by accident. After all, if we were all proper drivers and followed proper procedures, there would be no need for seat belts or air bags.

Summary

Process control systems are active and dynamic, therefore most faults are inherently self-revealing. Safety systems are passive or dormant, therefore many faults are *not* self-revealing. Safety systems require either manual testing or effective self-diagnostics, something many general purpose control systems do not incorporate very effectively.

Control systems are designed to allow relatively easy access so operators can make the frequent changes that are required. Safety systems, however, require strict security procedures and access control in order to prevent inadvertent changes. The control system, or any changes made to it, must not prevent the safety system from functioning properly.

In the not-too-distant past, process control and safety systems were implemented in separate, diverse technology systems. Pneumatic control systems are being replaced by software-based DCSs. Relay-based safety systems are being replaced by software-based PLCs. Incorporating both in one combined system offers potential benefits of a single source of supply, simplified stores, maintenance and training, and possibly lower cost. Standards, guidelines, and recommended practices from numerous industries,

however, all *strongly* discourage such a practice. The issue has little to do with the reliability of modern control systems.

The design of safety systems requires following strict and conservative design requirements. Analysis, documentation, design, operation, maintenance, and management of change procedures all require extra effort for safety systems. Implementing both control and safety in one system means that the extra effort must be applied to the *entire* control system. This is simply not economically feasible for any reasonably sized system. Placing all your eggs in one basket is rarely a good idea. Everything fails, it's just a matter of when.

If an organization chooses to diverge from the standards, they better have a valid, documented reason for doing so. A simple thought to keep in the back of your mind is, "How will I justify this decision in court?" If anything were to ever happen, that's what it may come down to. If your answer for violating the standards is, "It was cheaper that way", the court may not respond favorably.

References

1. Kletz, Trevor A. *Computer Control and Human Error.* Gulf Publishing Co., 1995.
2. *Programmable Electronic Systems in Safety Related Applications, Part 1 - An Introductory Guide.,* U.K. Health & Safety Executive, 1987.
3. *Guidelines for Safe Automation of Chemical Processes.* American Institute of Chemical Engineers - Center for Chemical Process Safety, 1993.
4. IEC 61508-1998. *Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems.*
5. ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod). *Functional Safety: Safety Instrumented Systems for the Process Industry Sector* and ISA-84.01-1996. *Application of Safety Instrumented Systems for the Process Industries.*
6. API RP 14C-2001. *Analysis, Design, Installation, and Testing of Basic Surface Safety Systems for Offshore Production Platforms.*
7. API RP 554-1995. *Process Instrumentation and Control.*
8. NFPA 85-2001. *Boiler and Combustion Systems Hazards Code.*
9. IEEE 603-1991. *Standard Criteria for Safety Systems for Nuclear Power Generating Stations.*

10. Smith, David J. *Reliability, Maintainability and Risk: Practical Methods for Engineers.* 5th Edition. Butterworth-Heinemann, 1997.
11. Neumann, Peter. G. *Computer Related Risks.* Addison-Wesley, 1995.
12. ANSI/ISA-91.00.01-2001. *Identification of Emergency Shutdown Systems that are Critical to Maintaining Safety in Process Industries.*